Design — Black-box replica (Challenge 2)

Problem

- Undocumented HTTP service: `POST /msg` with `{"m": "CMD;arg…;control"}`, stateful per token, plus `/__reset` and `/health`.
- Deliver an exact replica (same status + body, errors included), a spec where every claim cites the experiment log, and the log itself.
- Graded by 100 hidden sequences (40 single, 45 of 2–5, 15 of 6–12) against the **20:00 version**; the service may change during the day.

Assumptions

- **Verified in recon:** deterministic after reset (20 examples replayed twice, 0 diffs, log 148–188); unknown route → 404 `E404` before auth (192–194, 204, 208); bad/no token → 401 `E001`, including `/health` (201–203, 205–207); body errors `E100` / `E102` / `E103` (195–198); `X-Req` only on authorized `/msg`, never reset (not compared, ignored). Control digit is mod 97 over the text read in base 10, case-insensitive; one valid value per message, checked before account existence (325, 429–450, 548, 821, 914, 930, 946).
- **Still assumed:** exact letter/separator mapping of the control digit; `AC-/OP-` counters restart on reset; only the team token is valid in the replica.

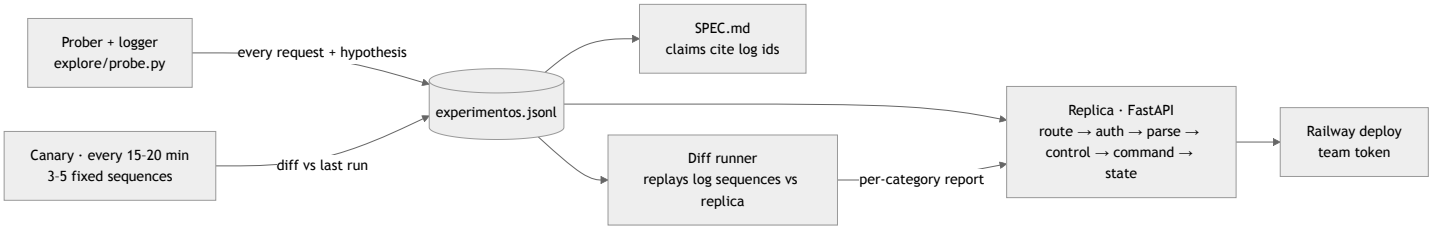
Hard cases

- Control-digit algorithm (`E201` malformed vs `E202` invalid) — blocks every other test.
- Error precedence when one message has several faults (control, arity, unknown command, missing account).
- `E506` limit: per operation or cumulative? `ANULA` side effects; odd amounts (0, negative, decimals, leading zeros).
- Hidden commands beyond the 6 seen; case-insensitive commands (`CONSULTa` works).
- Mid-day behavior change (incident announced for 15:30).

Out of scope

- Response headers (not compared) · concurrency / multiple instances (graders hit one process) · persistence across restarts (reset before each test) · replicating rate limits (infrastructure, not service).

Architecture



Decisions and trade-offs

Decision	Rejected alternative	Why
Deterministic replica, no LLM at runtime	LLM answering in runtime	Exact match tolerates zero variance; $0 per test; < 50 ms
Experiment log is also the test set (offline replay)	Live differential calls on every test run	Saves the 8k/day budget; repeatable; same file feeds spec evidence
Control digit: sweep 00–99 on ~20 messages, fit formula offline	Guess formulas from 20 examples	Byte-level guesses (sum, mod 97 over bytes, CRC32, XOR) failed; the sweep showed mod 97 over the text read as a base-10 number; sweep costs ~2k requests
Scripted exploration, hypotheses drafted with Claude Code	Autonomous LLM explorer (pi)	Faster to build and to control in 7 h; each request keeps an explicit hypothesis
Railway, single instance	Render free tier	No sleep before grading; one process keeps state consistent

Failure modes

- Undiscovered behavior → explicit default, marked **assumed** in the spec.
- Service changes → canary diff flags it; re-explore only the affected area; redeploy.
- False claims in the spec cost points → **confirmed** only with ≥ 2 log ids.
- Rate limit (`X-Frontier-Infra`) → excluded from evidence; prober stays < 10 rps.

Evaluation plan

Category	Test cases	Metric	Target
Recorded examples	the 20 from `ejemplos.jsonl`	exact match	100%
Single requests	entrada, auth, integridad, each command, each error	exact match per area	≥ 95%
Short sequences (2–5)	create → deposit → transfer → query; ANULA twice	exact match per sequence	≥ 90%
Long sequences (6–12)	limits, balances after ANULA, error precedence	exact match per sequence	≥ 85%
Hold-out	random sequences from a generator, never used for discovery	exact match	≥ 80%

- One command (`python evals/run.py`) prints per-category results; report before/after each change and after the mid-day change. Official score from `/evento/resultados` after each deploy.

Cost and latency

- Runtime: $0 LLM, < 50 ms per step (limit 5 s).
- Request budget (8k/day/token): recon ~210 · control sweep ~2k · exploration ~3k · canary ~1k · hold-out recording ~500 · reserve ~1.3k. `probe.py` prints used/remaining after each batch.
- LLM spend only for analysis with Claude Code; nothing from the $45 OpenRouter budget planned.